

ReDataX - Model Evolution

Model Evolution and Experimental Evidence

This dashboard separates two predictive experiments: **Local OOS**: M0, M1 and the M1R diagnostic **Cross-market final test**: M1, M2 and M3

Absolute Average Precision values should only be compared within the same experiment. The primary evidence is the paired daily AP uplift.

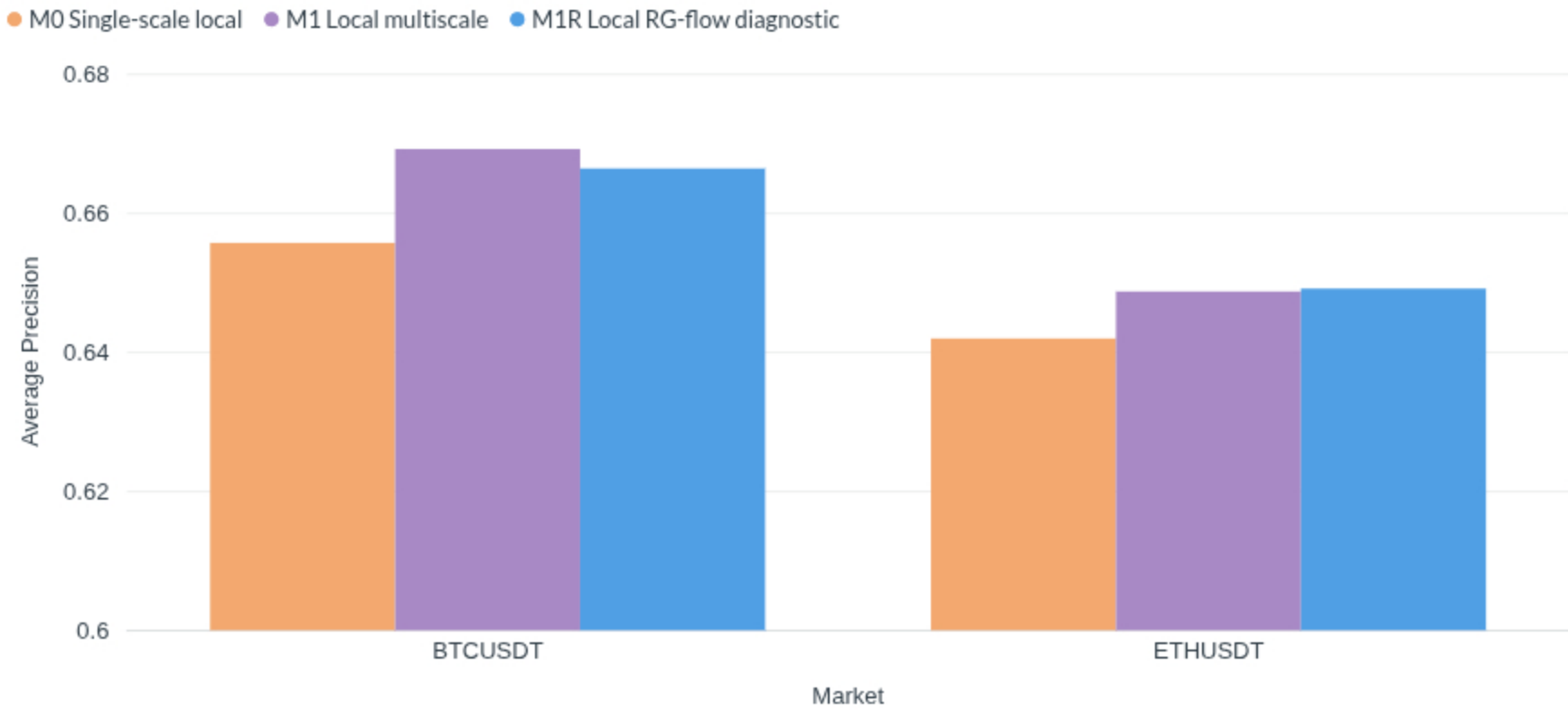
M1R is a local RG-flow diagnostic and is not the final cross-market M2.

Models

Model	Name	Family	Target	Status	Predecessor	Description
M0	Single-scale local baseline	classification	binary adverse-selection label	Baseline		Local single-scale predictive baseline.
M1	Local multiscale	classification	binary adverse-selection label	Accepted	M0	Local temporal coarse-graining across multiple scales.
M2	Cross-market RG-noJ	classification	binary adverse-selection label	Accepted	M1	Cross-market multiscale state without explicit pairwise J terms.
M3	Cross-market RG-with-J	classification	binary adverse-selection label	Rejected	M2	Explicit pairwise interaction terms; Small positive BTC and small negative ETH effect; No consistent cross-market improvement
M4	Direct value regression	regression	positive markout in bps	Accepted	M2	Direct prediction of positive future markout.
M5	Hurdle economic model	hurdle	probability and severity of positive markout	Final candidate	M4	Probability, conditional severity and break-even gate.

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Local OOS Experiment



Cross-market Experiment

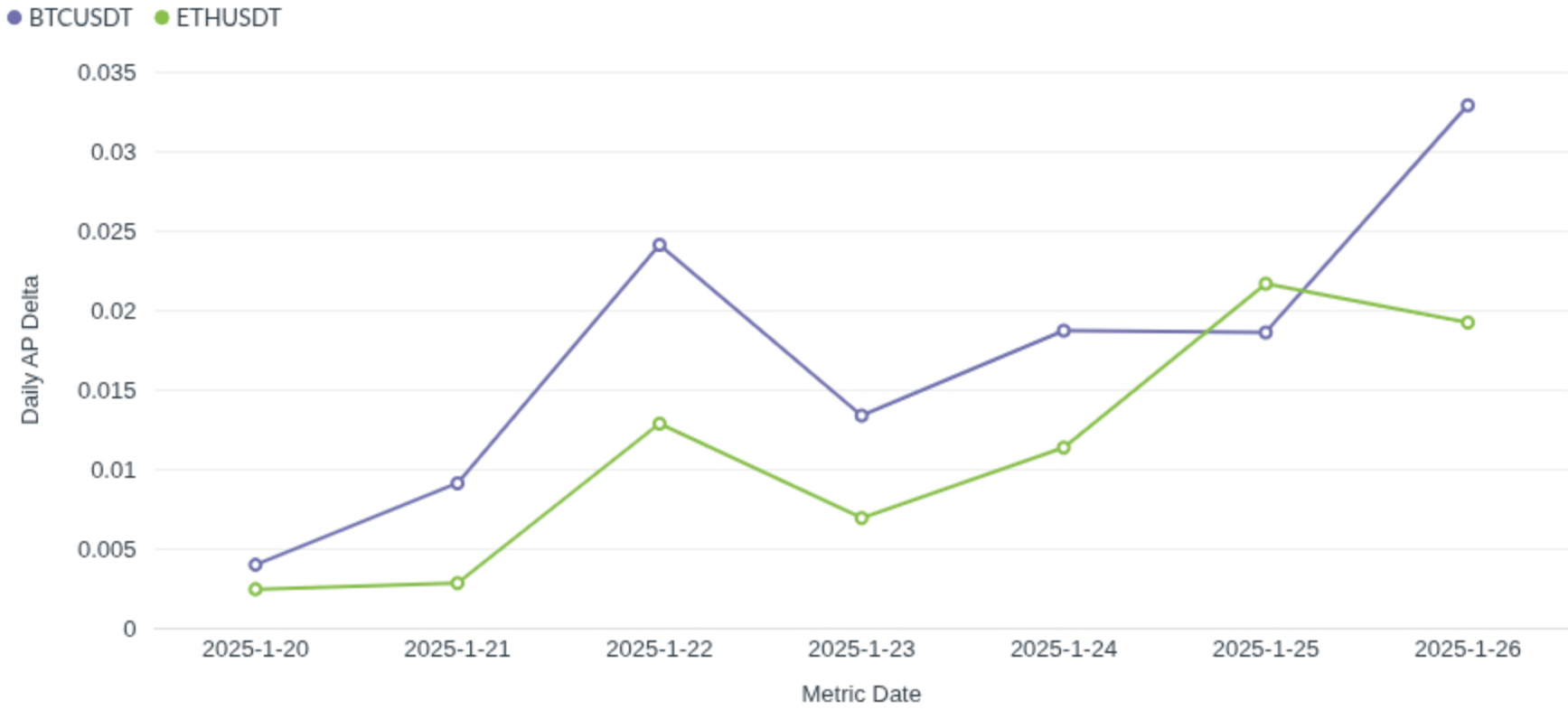


Incremental AP Uplift

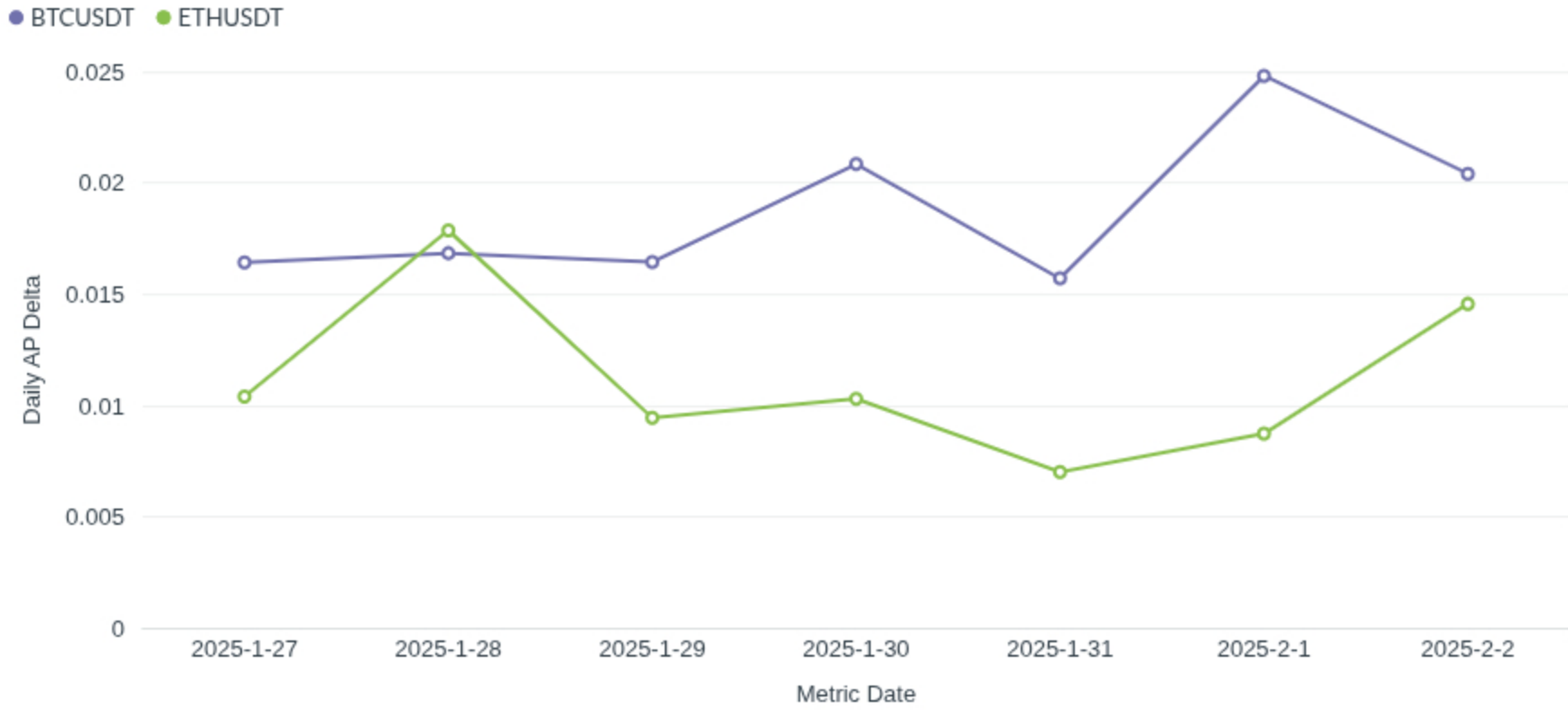
Market	Comparison	Mean Delta AP	95% CI	PositiveDays	Inference
BTCUSDT	M1 Local multiscale vs M0	0.017	[0.01095; 0.02406]	100%	Positive
BTCUSDT	M1R local RG-flow vs M1	-0.00051	[-0.00149; 0.00017]	57.1%	Not significant
BTCUSDT	M2 Cross-market RG-noJ vs M1	0.019	[0.01676; 0.02129]	100%	Positive
BTCUSDT	M3 RG-with-J vs M2	0.00026	[0.00011; 0.0004]	85.7%	Positive
ETHUSDT	M1 Local multiscale vs M0	0.011	[0.00597; 0.01647]	100%	Positive
ETHUSDT	M1R local RG-flow vs M1	0.00035	[-0.00009; 0.00071]	71.4%	Not significant
ETHUSDT	M2 Cross-market RG-noJ vs M1	0.011	[0.00892; 0.01385]	100%	Positive
ETHUSDT	M3 RG-with-J vs M2	-0.00023	[-0.00043; -0.00003]	14.3%	Negative

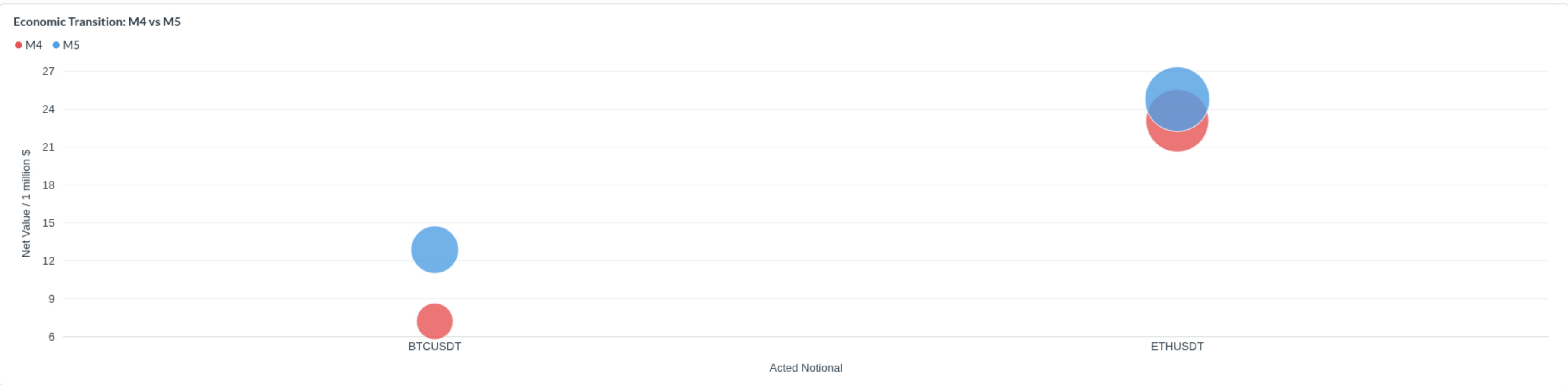
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Daily AP Uplift: M1 Compare M0



Daily AP Uplift: M2 Compare M1





Hypothesis Decision				
Hypothesis Order	Hypothesis	Evidence	Decision	
1	Temporal multiscale features improve the local baseline	M1 vs M0: positive paired AP uplift on the 5-second OOS test	Accepted	
2	The local RG-flow diagnostic adds value beyond multiscale features	M1R vs M1: near-zero and unstable incremental AP	Rejected	
3	Cross-market multiscale state improves the local model	M2 vs M1: positive AP uplift with 7/7 positive days	Accepted	
4	Explicit pairwise J terms improve the cross-market state	M3 vs M2: incremental AP is small and unstable	Rejected	
5	A 5-second economic policy covers the assumed action cost	The direct value model selected no robust profitable actions	Rejected / reformulated	
6	Longer markout horizons contain sufficient economic headroom	Oracle scan is positive on 120-600 second horizons	Accepted	
7	An economically constrained hurdle policy can produce positive out-of-sample value	P3 is positive VS No Action on both final holdout	Accepted with qualification	

Historical Ranking To Capture Evidence					
Market	Model	Event Capacity	Selected Notional	Exposure Capture	Loss Lift
BTCUSDT	M0 Single-scale local	10%	0.04%	0%	0
BTCUSDT	M1 Local multiscale	10%	6.79%	7.1%	0.71
ETHUSDT	M0 Single-scale local	10%	0.04%	0%	0
ETHUSDT	M1 Local multiscale	10%	0.38%	0.5%	0.05
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This historical experiment constrained the number of selected events, not selected notional. It motivated the later transition to explicit notional budgets in M4 and M5.